Nicholas Tang

tangnicholas26@gmail.com — +1 (925) 660-5921 — SF Bay Area, CA
Portfolio — GitHub — LinkedIn

Summary

Embedded firmware engineer with hands-on experience developing safety-critical systems in C/C++ for electric vehicles. Built STM32 firmware for PID-based traction control and regenerative braking, with robust communication over CAN and UART. Strong focus on low-level systems, real-time constraints, and hardware–software integration.

Education

UC Santa Cruz — Sept 2024 – Jun 2027
B.S. Computer Science, EE Minor — GPA: 3.9

Experience

(Incoming) Rivian and Volkswagen Group Technologies — June 2026 – Sept 2026
Software Engineering Intern, Charging

- Incoming software engineering intern on the Charging team

Formula Slug (Formula Society of Automotive Engineers) @ UC Santa Cruz — Sept 2024 – Present
Firmware Engineer (C++)

- Developing firmware for a custom electric vehicle for the FSAE international collegiate competition.
- Implemented automatic lap counting using GPS data from CAN bus with a flat-Earth approximation and tolerance of $\pm$ 10 meters.
- Managed 10 MHz UART communication with VN-200 for performant IMU and GPS data acquisition.
- Developed firmware for regenerative braking and PID-based traction control on STM32 MCUs.
- Utilized MbedOS RTOS, CMake, and Ninja for building and flashing to ARM-Cortex M cores.
- Engineered safety-critical power delivery systems, implementing LV undervolting detection and relay actuation.

UCSC Baskin Engineering (ECE) — Mar 2026 – Present
High Performance Power Electronics Lab

- Working on high performance power electronics hardware and associated embedded systems.
- Developing PCB in Altium for custom minimal TI C2000 modular breakout board.
- Responsible for sourcing parts.

UCSC Earth and Planetary Sciences — Jun 2025 – Jan 2026
Planetary Cloud Tracking Research (C/C++)

- Exploring computer vision algorithms to track wind patterns on Jupiter and other planetary atmospheres.
- Exposure to and translation between HDF5, NetCDF, TIFF, etc.
- Using CMake to manage complex, large-scale projects
- Implementing image/signal processing algorithms in C
- Solving Euler-Lagrange equations with numerical integration

Projects & Other Experience

NASA's Professional Development Program (NPWEE) — June 2025 – Aug 2025
Lead Systems Engineer

- Spearheaded design and writing for the final proposal of 7 pages.
- Researched airspace management and aviation systems; workforce development program.

Musical Auto-Transcribe DSP — Dec 2025 – Present

- Turns audio files into human readable musical notation.
- Applying FFT algorithms, Hann windowing, and general signal denoising.

Skills

Languages:	C/C++, Python, Java, C#, Node.js, Bash
Embedded/HW:	RTOS (MbedOS), STM32, CAN bus, UART, SPI, KiCad, Oscilloscopes
Tools/DevOps:	Git, CMake, Ninja, Docker, Valgrind, GNU Make, GDB, Vim
Math/Theory:	DSP, FFT, Numerical Integration, ODE Solvers, PID Control
AI/Productivity:	LLM Prompt Engineering (Copilot, Gemini, Grok), Technical Documentation